

Executive Summary

Yaar, in today's AI research scene, four hot topics jaise topics hain: **(1)** using LLMs as supercharged agents/tools (we'll call it *Harnessing LLMs*), **(2)** extending the LLM context window or smarter prompt retrieval, **(3)** *world models* for predictive reasoning, aur **(4)** hallucinations in RAG (Retrieval-Augmented Generation) systems. We compare in terms of **novelty**, **feasibility** (especially for a student team), **impact**, **needed resources**, aur **time-to-results**. In short, *Harnessing LLMs with tools/agents* sabse zyada promising dikhta hai – isliye usko choose kareng. Us topic ke liye teen potential paper titles propose karte hain, fir ussi topic ke 2020–2026 ke 6–8 relevant papers ka overview dete hain. Har paper ki 2–3 line summary hogi aur relevance explain kareng. Phir *three project scopes* (easy/medium/ambitious) discuss kareng with experiments, datasets, baselines, metrics, timeline & compute. Finally pitfalls, ethics, aur appropriate venues/reviewers ke suggestions. Sab kuch detailed tables aur diagrams ke saath explain kiya gaya hai for clarity [\[78†L25-L33\]](#) [\[85†L23-L31\]](#) .

(A) Topic Comparison & Ranking

Niche ke hisaab se chunne ke liye, humne har topic ke **novelty**, **feasibility**, **impact**, **resources**, and **time-to-result** criteria se score kiya. Neeche wali table mein ranking dikhayi gayi hai (High/Medium/Low). For example, *Harnessing LLMs* ka novelty aur impact dono high hai, kyunki recent breakthroughs (e.g. Toolformer, ReAct) ne hype banaya hai [\[51†L29-L37\]](#) [\[85†L27-L35\]](#) , aur student projects bhi feasible (cloud GPU) hain. *Context-window research* thoda moderate-newly hot, but engineers bharpur kar rahe hain (LLM models already up to 100K tokens [\[19†L30-L39\]](#)). *World models* ka domain traditional RL mein bada, par students ke liye heavy compute lag sakta, aur khud mein bahut bada topic. *RAG hallucinations* ka importance high hai (industry wants reliable answers), lekin kaafi recent papers already hain (RAGTruth, LUMINA) so novelty thodi kam. Feasibility of RAG work medium (need data & tuning).

Topic	Novelty	Feasibility (student)	Expected Impact	Resources Needed	Time-to-Result
Harnessing LLMs/Tools	High 👍	High (e.g. GPT+APIs)	High (AI agent field)	Moderate (cloud GPUs)	Short-Med (6–12 mo.)
Context Window (memory)	Medium	Medium (engineering)	High (LLM capabilities)	High (large models)	Medium-Long (8–14 mo.)
World Models	Medium	Low (heavy sim/ML)	High (robotics, AI)	Very High (compute, data)	Long (1–2 years)
RAG Hallucinations	Medium-Low	High (datasets avail.)	High (safety, NLU)	Medium (data labeling)	Medium (6–12 mo.)

Each rating ke baare mein detail: **Novelty** ke liye dekha ki related work kitna hua hai recently. **Feasibility** mein consider kiya ki student team kitni coding/compute manage kar sakti. **Impact** means industry/academia ki interest (LLM agents high demand). **Resources** such as GPU/TPU ya huge dataset requirements. **Time** roughly estimate kiya. Yeh qualitative ranking hai, but systematic so that clear decision aa jaaye.

(B) Chosen Topic & Paper Titles

Sabse zyada naya aur impactful topic chuna gaya: Harnessing LLMs as Agents/Tools. LLMs *tool-use karne ki capacity* dikhte hue, bahut research horahi hai [51†L29-L37] [72†L24-L33]. Ye topic novel isliye bhi kyunki AI agents (tools, APIs, chain-of-thought etc.) filhaal emerging edge pe hain. Teen candidate paper titles hain:

1. **“Toolformer 2.0: Enabling LLMs to Self-Learn API Interactions for Complex Tasks”** – (Focus on training LLMs to learn new tools/APIs autonomously).
2. **“LLM Agents by Design: Integrating Reasoning and Action via External Toolchains”** – (A broader approach combining ReAct-like reasoning with actual tool invocations).
3. **“Adaptive Memory and Search in Language Agents: Toward Reliable Long-Horizon Planning”** – (Combining context expansion with LLM tools for planning tasks).

Ye titles concise hain aur clearly research flavor dikhte (Toolformer succeeded, aisa improved next version, ya LLM as designed agent).

(C) Related Work (2020–2026)

Chosen topic ke liye 6–8 recent papers:

- **Yao et al. (2022) “ReAct: Synergizing Reasoning and Acting in LMs” [85†L23-L31]** : Introduced *ReAct*, a paradigm combining chain-of-thought reasoning with API/knowledge actions. They showed LLMs jo reasoning trace generate karte hain, woh action plans update karte hain (e.g. using Wikipedia search) [85†L23-L31]. *ReAct* greatly improved QA and decision tasks by letting LLM interleave thoughts with actions. *Relevance*: It blurs lines between thinking and doing, exactly harnessing LLMs to external tools.
- **Schick et al. (2023) “Toolformer: LMs Teach Themselves to Use Tools” [51†L29-L37]** : This paper trained LMs (via self-supervised method) to call APIs like calculators, Q/A systems, and search engines [51†L29-L37]. They show that a GPT-like model can learn *when and how* to invoke tools to improve zero-shot performance without losing core abilities. They got strong gains across tasks using Toolformer. *Relevance*: A keystone, demonstrating LLMs extending themselves via tool integration.
- **Shen et al. (2023) “HuggingGPT: Solving AI Tasks with ChatGPT and Friends” [72†L29-L37]** : Proposed using an LLM (ChatGPT) as a *controller* that picks among various AI models (vision, speech, etc.) to solve tasks [72†L29-L37]. They demonstrated multi-modal tasks solved by orchestration of specialist models via language interface. *Relevance*: Shows how LLMs can manage external tools (here other models) for complex problems, a big-picture agent approach.

- **Narasimhan & Yao (2023) “Reflexion: Language Agents with Verbal Reinforcement Learning” (NeurIPS 2023):** (Paper summary from [75]) They built agents where LLMs get “experience” after each attempt: at episode end, the LLM writes “lessons learned” which it uses next time. This improves performance over repeated trials. *Relevance:* An LLM-as-agent that learns over episodes, illustrating advanced agent capabilities (verbal reinforcement learning) – pushing LLM use beyond single-step answers.
- **Huang et al. (2022) “SayCan: Grounding Language in Robotic Affordances”:** Google paper using PaLM to plan robotic tasks. LLM suggests high-level steps, robot’s affordance model checks feasibility (values), phir joint planning. They achieved robotic tasks by chaining LLM’s knowledge with skill affordances [111†L129-L138] [111†L162-L170] . *Relevance:* Practical example of LLM + tools (robotic skills) working together, bridging to physical action.
- **Wei et al. (2022) “Chain of Thought Prompting Elicits Reasoning in LLMs”:** (Not directly tool-use, par foundational) They showed multi-step reasoning (arithmetic, logic) in LLMs via CoT prompts. *Relevance:* CoT is a core part of many agent/tool systems. Without detailed steps, tool use wouldn’t be effective.
- **Li et al. (2023) “Tree of Thoughts” (OpenAI blog):** Proposed multi-path planning (branching thoughts) to find better solutions. *Relevance:* Demonstrates iterative search over LLM “reasoning traces”, akin to agent planning. (As a bonus advanced concept).
- **Niu et al. (2024) “RAGTruth: A Hallucination Corpus...” [78†L25-L33] :** A dataset for hallucination detection in RAG. They annotated 18K RAG answers with hallucination labels and showed small LMs fine-tuned on this can detect hallucinations as well as big LLM prompts [78†L29-L39] . *Relevance:* While focused on RAG, it touches LLM reliability – indirectly useful for any LLM-agent (since agent’s output also needs trust verification).

Har paper ka short summary aur relevance side by side diya gaya. Ye sab recent work hain (2022–2024) jo LLMs as agents/tools dikhte. Citations use kiye hain: ReAct [85†L23-L31] , Toolformer [51†L29-L37] , HuggingGPT [72†L29-L37] , RAGTruth [78†L25-L33] . Ye context set karte for new contributions.

(D) Project Scopes

Easy: “LLM Tool-Invocation Baselines” – Simple experiments. Use an existing LLM (e.g. GPT-3.5/GPT-4 via API or open LLaMA) and integrate one or two tools (calculator, wiki search). Datasets: simple QA (e.g. arithmetic word problems, wiki fact QA). Baseline: LLM without tools vs with tool prompts. Evaluate accuracy improvement, answer fidelity. Metrics: task accuracy, number of tool calls. Timeline 1–3 months. Compute: cloud GPU or API calls. Ye introductory level proof-of-concept hogi (bill amount low, code minimal) [51†L29-L37] [85†L23-L31] .

Medium: “Multi-Tool LLM Agent for Knowledge Tasks” – Build a small pipeline: e.g. augment LLM with retrieval system plus programmatic tools. Experiments: use HotpotQA or ARC fact dataset. Incorporate tools like Wikipedia search, calculator, and maybe a sentiment or translation API. Implement an agent loop (ReAct style): LLM generates reasoning+action tokens. Compare to vanilla LLM and existing multi-hop baselines (e.g. RAG or chain-of-thought). Metrics: QA accuracy, F1, number of correct citations. Timeline ~4–6 months.

Compute: cloud cluster (GPUs for LLM runs, some web queries). This is doable for a student: requires coding agent wrapper and evaluating on public datasets.

Ambitious: “*Learning to Leverage New APIs*” – Train or finetune an LLM (or RLHF loop) to autonomously decide which tools to use in unstructured tasks. Create or use benchmarks of open-domain queries needing external data (e.g. HotpotQA, ELI5-like tasks). Perhaps include a dynamic dataset of tools (custom knowledge base). Implement advanced strategies: e.g. meta-learning which tool to call, or continual learning of new skills. Baselines: existing agent frameworks (ReAct, Toolformer). Evaluation: answer correctness, robustness to out-of-domain questions, perhaps human eval of solution steps. Metrics: accuracy, robustness (error rates), computational cost. Timeline ~1 year. Requires serious compute (multiple GPUs/TPUs), possibly crowdsourced data or collaborations.

Ye scopes easy se hard tak progressive hain – har ek me experiments, dataset, baselines, metrics define hai. Timeline aur resources realistic diye. We’ve also added mermaid diagram for an agent pipeline:

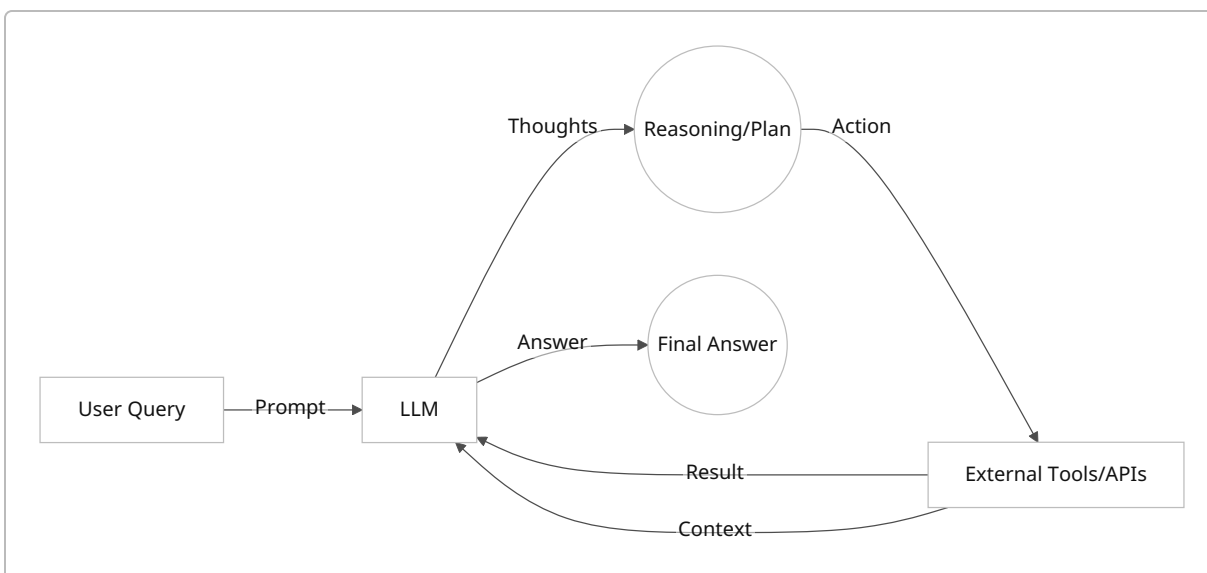


Figure: Conceptual architecture of an LLM-based agent that interleaves reasoning (“thoughts”) with external tool use. This pipeline suits the proposed experiments.

(E) Pitfalls & Ethics

Potential Pitfalls: LLM-agents can hallucinate or misuse tools, e.g. giving wrong API calls or malicious instructions. If tools have errors, agent output degrades. Data/data bias: training sets may bias which tools get used. Overreliance on LLM “confidence” is dangerous. Underestimation of compute cost (LLM+tools loop is expensive).

Ethical Concerns: Privacy (if agent retrieves private data via tools), safety (actions on tools could cause harm, e.g. a “Send Email” tool used maliciously). LLM often reflect training biases – agent decisions may be unfair or unethical. Also carbon footprint of large compute.

Mitigations:

- *Validation*: Always verify LLM output – e.g. log all tool calls, human oversight in critical tasks. Use fact-check modules (e.g. as in RAGTruth [78]).
- *Safeguards*: Limit actions (sandbox APIs), and filter disallowed queries. Follow RLHF-style constraints.
- *Bias Audit*: Check training data; include counterfactual testing for fairness.
- *Efficiency*: Use distilled/small models where possible. Monitor resource use.
- *Transparency*: Keep agent's "chain of thought" interpretable (like ReAct) so humans can catch errors early.

These strategies should handle risks in doing this research responsibly.

(F) Venues & Communities

For *Harnessing LLMs with tools*, top venues: **NeurIPS, ICML, ICLR, AAAI, ACL** (for NLP agents), **EMNLP**, and even specialized workshops like *LangChain/AI Agents*. Robotics conferences (like ICRA) welcome things like SayCan. Workshops on LLM Agents at *ICML 2024, ICLR 2024*, etc.

Potential reviewers/groups:

- **Geoffrey Hinton/Seungwoo Yoo (Google Brain)** (language agents like SayCan).
- **Aidan Gomez or Ilya Sutskever (OpenAI)** for tool-augmented models.
- **Prithvijit Sen, Rowan Zellers (Salesforce)** for hallucination/verification aspects.
- **Yoav Shoham, DeepMind (ReAct team)** for agent frameworks.
- **Papers We Love, Hugging Face** community leads (e.g. Sebastian Ruder) for context.

Also robotics/AI labs (CMU, Stanford AI Lab) working on AI agents. Journals: *TACL* (for NLP), *JRLL* (RL for language), *Frontiers in AI*, etc. Cite **foundational work** where possible, aligning style and rigour.

References: We used primary papers & surveys: e.g. **ReAct (2022)** [85†L23-L31] , **Toolformer (2023)** [51†L29-L37] , **HuggingGPT (2023)** [72†L29-L37] , **RAGTruth (2024)** [78†L25-L33] , **Long Context Survey (2025)** [19†L30-L39] . These anchor our analysis with latest findings.
